

1. Local Earth-System Controls

- Spatio-temporal rainfall dynamics
- Soil-moisture, groundwater, snow, urban infrastructure,
- Hydraulic connectivity, storage, and infrastructure

2. Scale Transition

- Aggregated climate forcing
- Smooth terrain and storage
- Omit unresolved processes
- What is lost, and what remains resolved?

3. Observe and Correct

- Remote sensing, gauges
- Socio-hydrology, cameras, sensors
- Data assimilation

4. Locally Relevant Outputs

- Flood Hazard
- Exposure and vulnerability
- Effective mitigation
- Cost-benefit analysis

